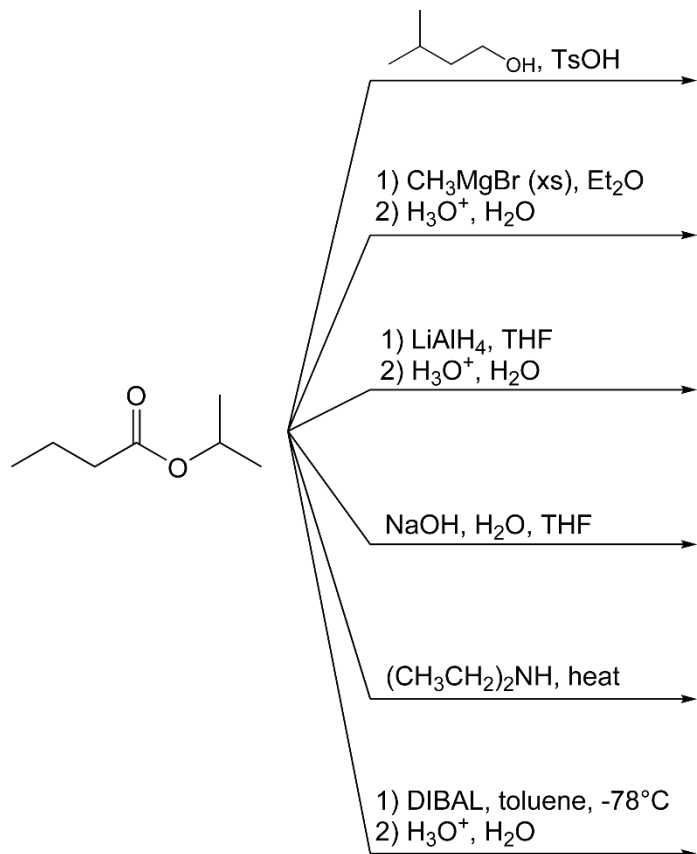
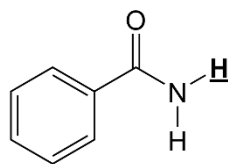
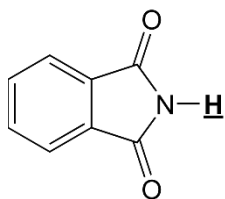


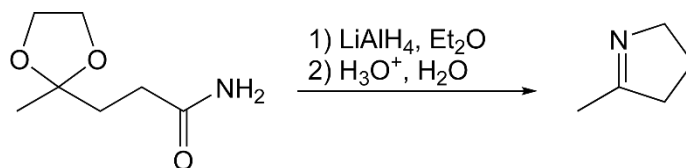
1. Give the products of the reaction of isopropyl butanoate in each of the following situations.



2. Phthalimide and benzamide are shown below. Predict which has a lower pKa and explain your choice.



3. When you treat the following amide with  $\text{LiAlH}_4$  followed by an acidic workup, the product below is isolated. Explain this observation.



4. Draw mechanisms for the following reactions.

